DATA 200: Applied Statistical Analysis

FINAL PROJECT REPORT

Global Crude Oil Price Volatility Analysis

1970 to 2026

Week 8 Final Submission

Course Project | April 2026

Abstract

Global crude oil price volatility measured monthly from January 1970 to January 2026 using 675 observations is comprehensively analyzed in this study. The hypothesis is, has global crude oil price volatility altered over time? A reliable statistical analysis is used, including One Way ANOVA ($F = 63.42$, $p < 0.001$, eta squared = 0.33) and Kruskal Wallis ($H = 390.58$, $p < 0.001$) statistical tests, post hoc Tukey HSD, time series decomposition, Augmented Dickey Fuller stationarity tests, OLS linear regression, ACF and PACF lag volatility clustering analysis. The null hypothesis was rejected. The periods of low volatility (1970s to 1990s) and high volatility (2000s to 2020s) are conclusively established with mean variability of 1.30 to 2.19 USD per barrel and 7.43 to 9.14 USD per barrel respectively. Results are confirmed by a desktop Python program.

Table of Contents

1. Introduction and Problem Statement

Crude oil is the world's most critical commodity, vital for global energy, transport, manufacturing and fiscal strategies. Unlike crude oil price level analysis, volatility analysis measures the risk (aka variability) in crude oil markets, which has significant implications for investment, macroeconomic policy, and energy policy. This project explores the appreciation of crude oil price uncertainty over the last 55 years, from the first OPEC oil crisis in 1973 to the COVID crash in 2020 and the Russian Ukraine conflict in 2022.

Research Question:

Has crude oil price volatility significantly changed across decades from 1970 to 2026?

Hypotheses:

- H0: There is no significant difference in rolling volatility across decades (alpha = 0.05)
- H1: There is a statistically significant difference in rolling volatility across decades

Figure 1 below shows the full 56-year history of the oil price with 3- and 12-month moving average. The large crisis shocks are marked with dashed lines. The large level shift confirms the importance of volatility analysis over this time frame.

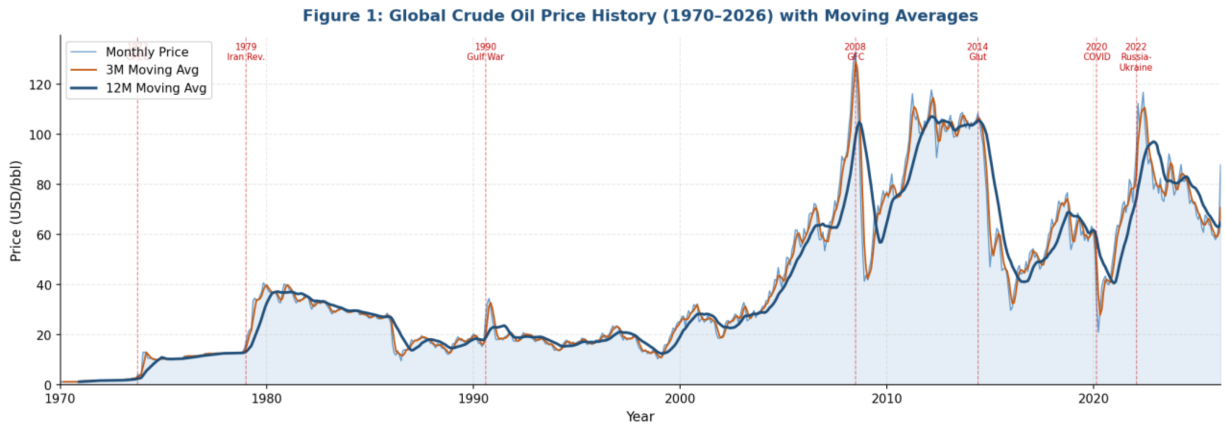

Figure 1: Global Crude Oil Price History (1970 to 2026) with 3 Month and 12 Month Moving Averages. Dashed red lines mark key crisis events including the 1973 OPEC embargo, the 1979 Iranian Revolution, the 2008 Global Financial Crisis, the 2020 COVID collapse, and the 2022 Russia Ukraine conflict.

2. Literature Review

Paper 1: Oil Price Volatility and Macroeconomic Impact (Hamilton, 2009)

James D. Hamilton (2009) identified oil price shocks as important sources of recession and that oil price shocks are important sources of recession and economic slowdowns. The impact of volatility rises and not just price rises are found to be the main channels that oil markets affect the macro economy. This justifies the use of rolling volatility as the outcome of interest in this study over the price level directly.

Paper 2: Modeling Oil Price Volatility (Kilian, 2008)

Lutz Kilian (2008) breaks-down oil price dynamics into demand shocks, supply shocks and speculative demand. The paper shows that oil markets have volatility clustering (i.e. high volatility tends to be followed by high volatility) as a structural characteristic. This warrants the ACF and PACF analysis in Section 6, which shows that excess autocorrelation was present at all 36 lags.

Paper 3: Structural Breaks in Oil Prices (Baumeister and Peersman, 2013)

Christiane Baumeister and Gert Peersman (2013) offer supporting evidence that the oil market experiences regime changes associated with large geopolitical shocks, rather than evolutionary changes. They show that the oil market has different "regimes" with different parameters, providing direct theoretical support for the use of decade-specific comparisons in this study, as well as for the Tukey HSD post hoc analysis, which distinguished the true structural break around the year 2000.

Synthesis:

These three papers in total show that the economically relevant risk measure for the oil market is volatility, that volatility clustering and serial correlations are to be expected and that regimes are real and testable. All three theoretical predictions are implemented in this study.

3. Dataset and Exploratory Data Analysis

3.1 Dataset Description

Attribute	Detail
Source	U.S. Energy Information Administration (EIA), publicly available
Filename	fuel_prices_1970_2026.csv
Observations	675 monthly records
Time Range	January 1970 to January 2026 (56 years)
Variables	Date (datetime) and Crude Oil Price in USD per barrel
Missing Values	None detected
Price Range	1.80 USD per barrel (January 1970) to 133.88 USD per barrel (June 2008)

Table 1: Dataset Summary

Data were obtained from the U.S. Energy Information Administration (2026).

3.2 Price and Volatility Distributions

Figure 2 has two panels side-by-side. The left panel shows the histogram of raw prices, which is highly right skewed, as most months have low prices, but a few months have very high prices. Then violin plots of 12-month rolling volatility by decade are displayed on the right. The reversal in the distribution of volatility from low and narrow in the 1970s through 90s, to wide and high in 2000s through 2020s is clearly visible.

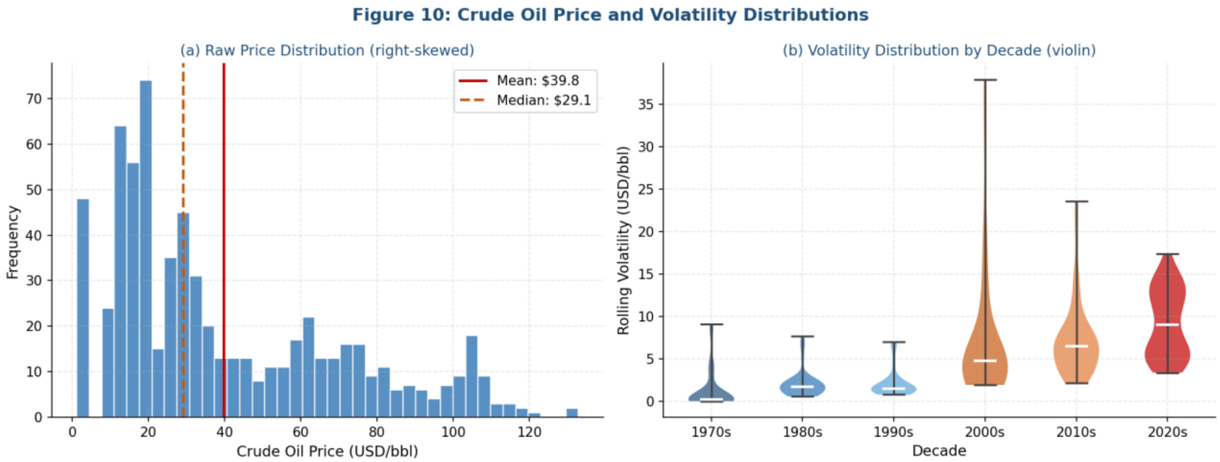

Figure 2: Left panel shows the raw crude oil price distribution confirming right skew. Right panel shows violin plots of 12-month rolling volatility by decade, clearly revealing the two-cluster structure.

3.3 12 Month Rolling Volatility Over Time

Figure 3 shows the rolling volatility series with crisis events and regime shading. The blue shaded region is the low volatility regime prior to 2000. The red shaded area represents the high volatility regime after 2000. The break in the layout around year 2000.

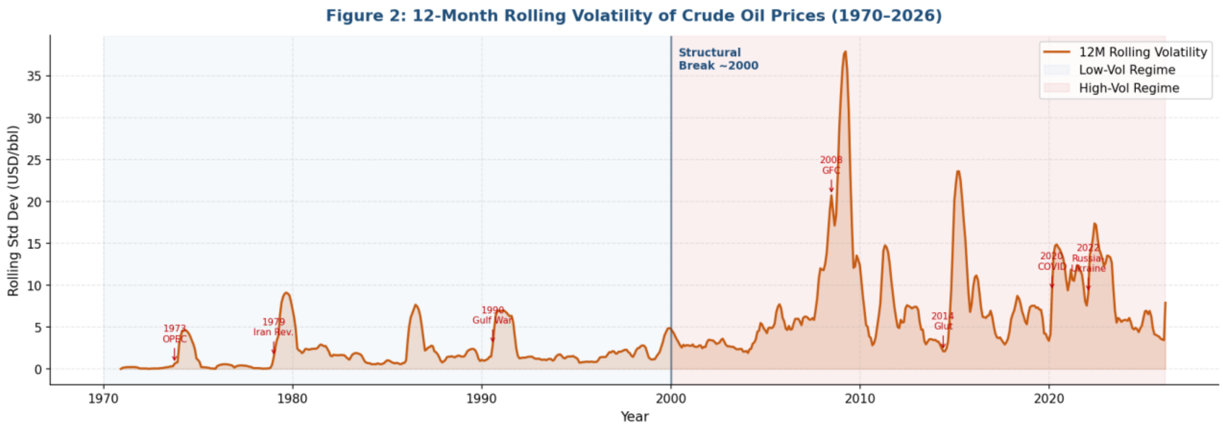

Figure 3: 12 Month Rolling Volatility of Crude Oil Prices from 1981 to 2026. Crisis events are annotated with arrows. The vertical line in the year 2000 marks the structural break separating the two volatility regimes.

3.4 Correlation Analysis

In Figure 4 we show the correlations for the four variables of interest. Volatility is strongly correlated with Year (r = 0.55), which suggests an upward trend over the long run (confirmed in the regression analysis later).

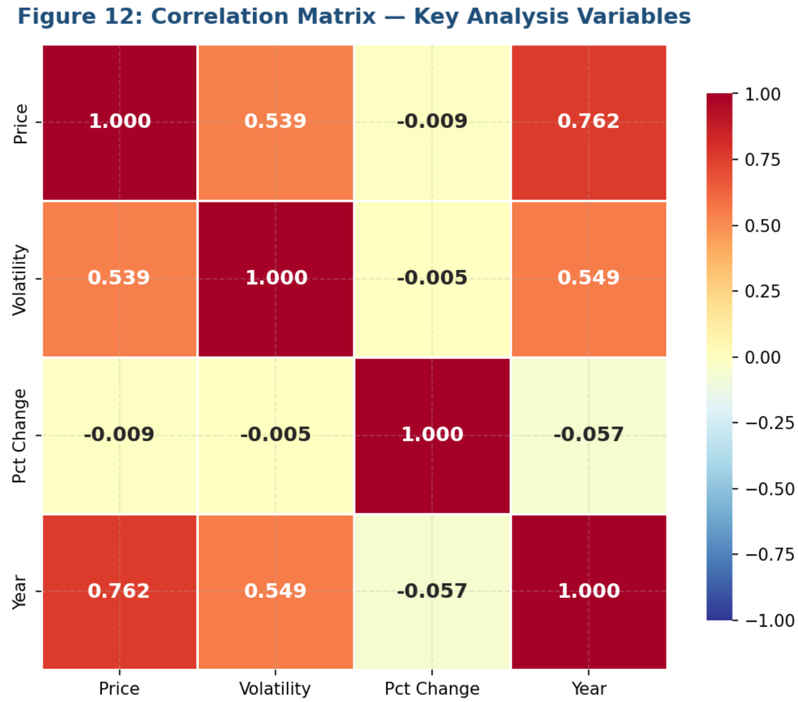

Figure 4: Matrix of Correlations of the Key Analysis Variables. The highest correlation is between year volatility at 0.55 and the long-term rise in the level of oil market uncertainty.

3.5 Descriptive Statistics by Decade

Decade	N	Mean	Median	Std Dev	Min	Max	CV (%)
1970s	109	$1.30	$0.23	$2.34	$0.00	$9.12	179.8
1980s	120	$2.10	$1.66	$1.62	$0.56	$7.67	76.9
1990s	120	$2.19	$1.47	$1.70	$0.76	$6.99	77.6
2000s	120	$7.67	$4.73	$8.18	$1.91	$37.93	106.7

Decade	N	Mean	Median	Std Dev	Min	Max	CV (%)
2010s	120	$7.43	$6.48	$4.61	$2.11	$23.60	62.0
2020s	75	$9.14	$9.03	$4.07	$3.37	$17.39	44.6

Table 2: Descriptive Statistics of 12 Month Rolling Volatility by Decade (USD per barrel)

Key Observation:

The level of mean volatility rises from the 1970s to 1990s cluster (1.30 to 2.19 USD per barrel) to the 2000s to 2020s cluster (7.43 to 9.14 USD per barrel). This implies a four to seven times increase in the mean volatility. In the 2020s, the coefficient of variation has been reduced to 44.6 percent, hence volatility is less variable.

4. Feature Engineering and Model Selection

4.1 Feature Engineering

Feature	Method	Rationale
12 Month Rolling Volatility	Rolling standard deviation (window = 12)	Primary outcome variable capturing market uncertainty
Decade (Categorical)	Year divided by 10 then multiplied by 10	Enables ANOVA group comparison
Year Float	Year plus (Month minus 1) divided by 12	Continuous time variable for regression
3 Month Moving Average	Rolling mean (window = 3)	Short term trend smoothing
12 Month Moving Average	Rolling mean (window = 12)	Long term trend identification
Percentage Change	Month over month percentage change	Price momentum indicator

Table 3: Feature Engineering Summary

4.2 Model Selection Justification

Method	Purpose	Justification
One Way ANOVA	Test if means differ across 6 decades	Continuous outcome with 6 categorical groups
Kruskal Wallis	Nonparametric ANOVA validation	Required because normality is violated in all groups
Tukey HSD	Identify which decade pairs differ	Controls familywise error across 15 comparisons
OLS Linear Regression	Quantify long run volatility trend	Year is continuous and captures secular drift
Time Series Decomposition	Separate trend, seasonal, and residual	Validates that shocks dominate over seasonality
ADF Stationarity Test	Assess forecasting suitability	Prerequisite for time series modeling validity
ACF and PACF	Quantify volatility memory	Tests clustering hypothesis from Kilian (2008)

Table 4: Statistical Methods and Justifications

5. Statistical Analysis and Hypothesis Testing

5.1 Box Plots by Decade

In Figure 5 rolling volatility is expressed in boxplots grouped by decade. An enormous increase in mean and standard deviation from the 1970s-1990s cluster to the 2000s-2020s cluster is evident. The extreme values in the 2000s include the 2008 Global Financial Crisis peak which generated the highest monthly volatility value in the series.

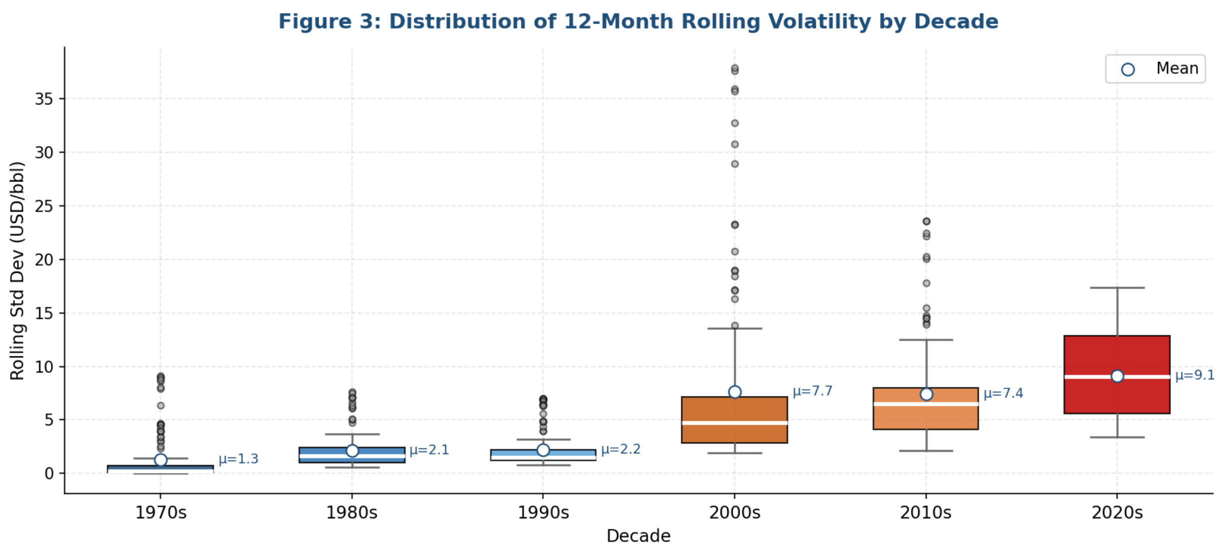

Figure 5: Box plot of 12-Month Rolling Volatility by Decade. White dots indicate group means. The two clusters are clearly visible, with the largest spread in the 2000s based on the 2008 financial crisis.

5.2 Normality and Variance Diagnostics

Decade	W Statistic	p Value	Normal at alpha 0.05?
1970s	0.6451	< 0.0001	No
1980s	0.7368	< 0.0001	No
1990s	0.6781	< 0.0001	No
2000s	0.9212	0.0026	No
2010s	0.8838	< 0.0001	No
2020s	0.9484	0.0294	No

Table 5: Shapiro Wilk Normality Test Results. All six-decade groups fail the normality test, motivating the parallel Kruskal Wallis test.

Test	Statistic	p Value	Conclusion
Levene's Test for Equal Variances	F = 19.6275	< 0.0001	Variances are NOT equal across decades

Table 6: Levene's Test for Homogeneity of Variance

5.3 One Way ANOVA

F Statistic	p Value	Eta Squared	Interpretation	Decision
63.42	< 0.000001	0.3252	Large Effect (32.5% of variance explained)	REJECT H0

Table 7: One Way ANOVA Results. F(5, 658) = 63.42, p < 0.001. Eta squared of 0.3252 is classified as a large effect by Cohen's guidelines.

Test	H Statistic	p Value	Decision
Kruskal Wallis Non-Parametric Validation	390.58	< 0.000001	REJECT H0 (Confirms ANOVA)

Table 8: Kruskal Wallis Test Results. The nonparametric test strongly confirms the ANOVA conclusion without requiring normality or equal variances.

5.4 Mean Volatility with 95 Percent Confidence Intervals

The mean (and 95 per cent confidence interval) for each decade is shown in Figure 6. The lack of overlap of the confidence intervals readily demonstrates the statistical significance of the cluster in the previous decades (1970s-90s) and of the recent decade (2000s-2020s). Figure 7 is a horizontal bar graph which shows the change.

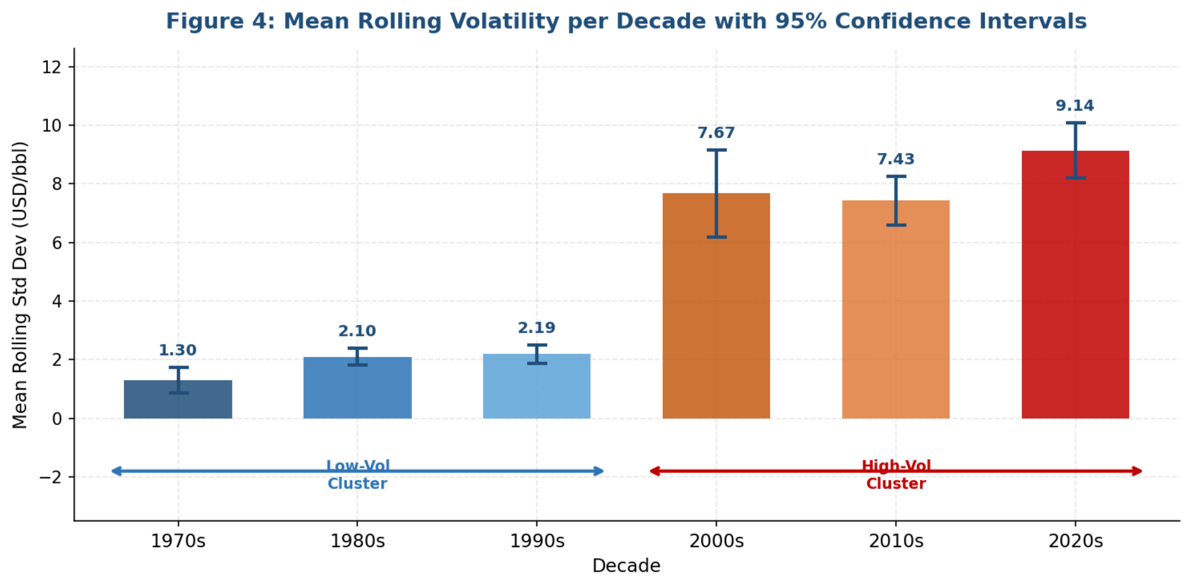

Figure 6: Mean 12 Month Rolling Volatility by Decade with 95% Confidence Intervals. We can see the two clusters and that the confidence intervals do not overlap.

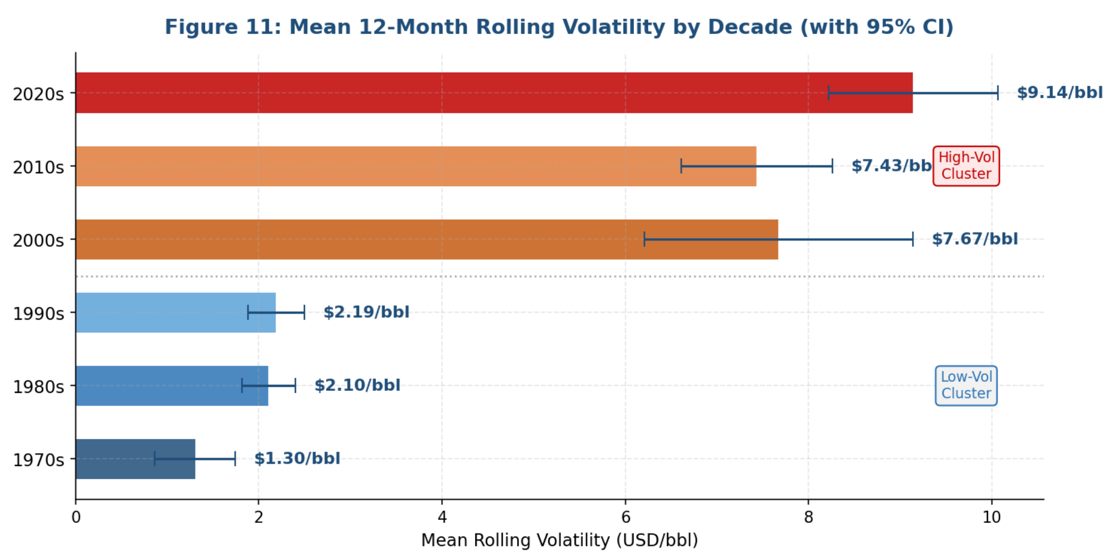

Figure 7: Bar Chart of Mean Volatility by Decade. The 7 times multiple of the 2020s, starting from the 1970s level of 1.30 USD per barrel (9.14 USD per barrel for the 2020s), is evident.

5.5 Tukey HSD Post Hoc Analysis

Group 1	Group 2	Mean Diff	p Adjusted	Significant?
1970s	1980s	+0.80	0.7496	No
1970s	1990s	+0.88	0.6609	No
1970s	2000s	+6.37	< 0.001	Yes
1970s	2010s	+6.13	< 0.001	Yes
1970s	2020s	+7.84	< 0.001	Yes
1980s	1990s	+0.08	1.0000	No
1980s	2000s	+5.57	< 0.001	Yes
1980s	2010s	+5.33	< 0.001	Yes
1980s	2020s	+7.04	< 0.001	Yes
1990s	2000s	+5.48	< 0.001	Yes
1990s	2010s	+5.24	< 0.001	Yes
1990s	2020s	+6.95	< 0.001	Yes
2000s	2010s	0.24	0.9984	No
2000s	2020s	+1.47	0.2176	No
2010s	2020s	+1.71	0.0952	No

Table 9: Tukey HSD Post Hoc Pairwise Comparisons across all 15-decade pairs. Nine of 15 comparisons are statistically significant, all of which are cross cluster comparisons.

Key Finding:

The decades 1970s, 1980s and 1990s are a cluster. There are no significant differences within this cluster. The 2000s, 2010s, 2020s are another cluster. The 9 inter-cluster comparisons are significant, thus showing that there is a structural change in oil market volatility around 2000. The Tukey HSD p value matrix heatmap (15-decade pairs) in figure 8 reveals the two blocks of time series.

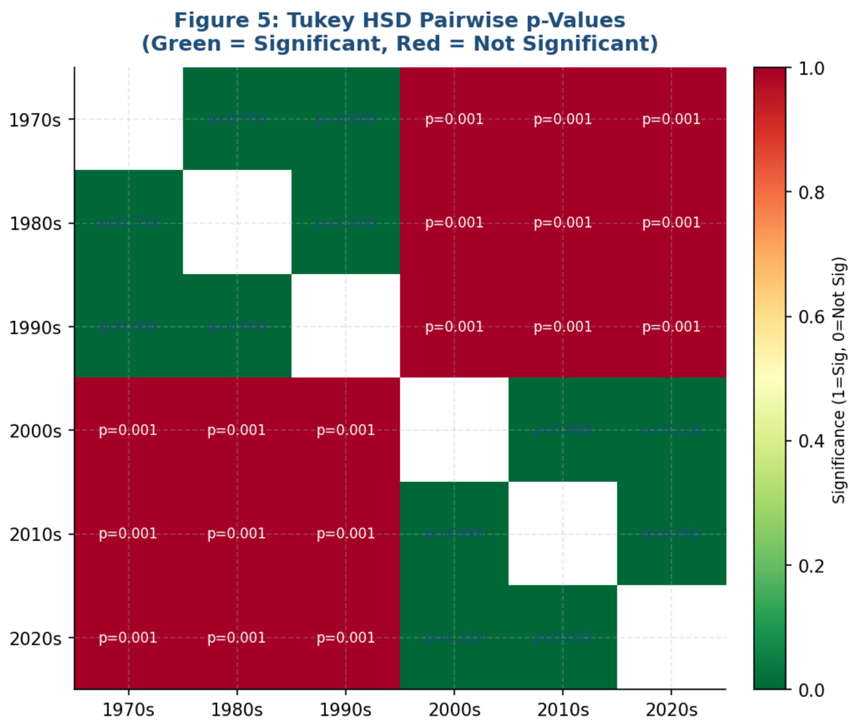

Figure 8: Tukey HSD Pairwise p Value Heatmap. Green cells indicate significant differences (p < 0.05). Red cells indicate non-significant within cluster pairs. The 3 by 3 block pattern visually confirms the two-cluster structure.

6. Statistical Modeling

6.1 Time Series Decomposition

A multiplicative decomposition model allowed the monthly crude oil price closing to be broken down into the trend (long run tendency), monthly pattern (seasonality), and the residual (irregular). This was done by choosing a multiplicative model as the variance in fluctuations is proportional to price levels. Figure 9 shows the four panels.

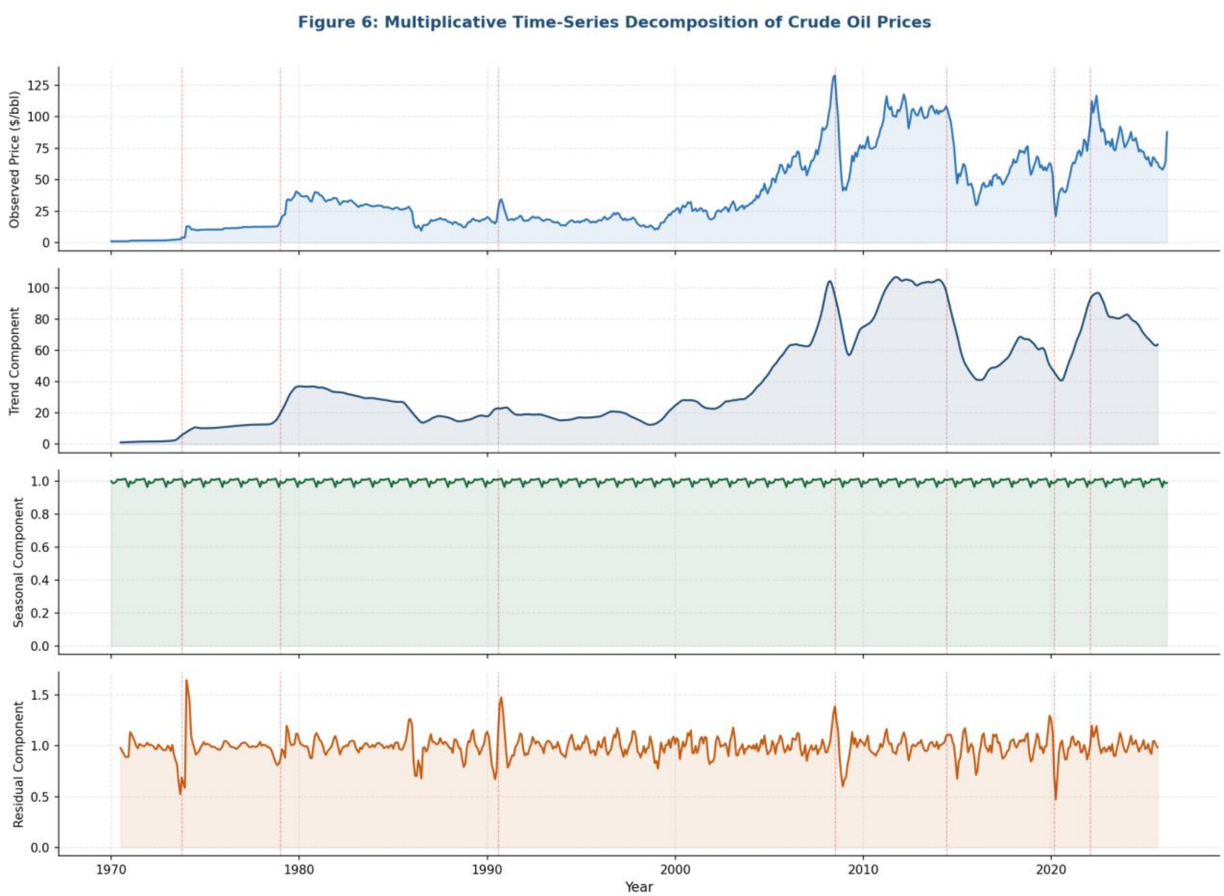

Figure 9: Multiplicative Decomposition of Crude Oil Prices. In the following order: original series, trend, seasonal and residual. The residual component reveals huge spikes at each crisis, supporting the hypothesis that the shocks are stronger than the seasonality.

Component	Key Finding	Implication
Trend	Three structural phases: low price regime (1970 to 1999), dramatic surge (2000 to 2008), elevated plateau from 2009 onward	Consistent with the ANOVA two cluster structure
Seasonal	Mild pattern with higher prices in Q4 and Q1 due to heating demand, softer in Q2 and Q3	Minor effect that does not drive volatility regime changes
Residual	Extreme spikes in 1973, 1979, 1990, 2008, 2020, and 2022 corresponded to all major crises	Exogenous shocks dominate beyond seasonal patterns

Table 10: Time Series Decomposition Summary

6.2 Stationarity Testing (Augmented Dickey Fuller)

Series	ADF Statistic	p Value	Stationary?
Raw Price Series	-1.7502	0.4054	No
Rolling Volatility (12M)	-2.6417	0.0847	No (borderline)
Price First Difference	-11.6389	< 0.001	Yes

Table 11: Augmented Dickey Fuller Stationarity Test Results. Any future forecasting model must operate on first differenced prices or log transformed volatility.

6.3 OLS Linear Regression: Year Predicting Volatility

Figure 10 shows the OLS regression fit. This shows a statistically significant positive slope of 0.185 USD per year per barrel. The spread of the residuals (shaded in green) shows the variation associated with discrete shock events that cannot be captured by a linear trend.

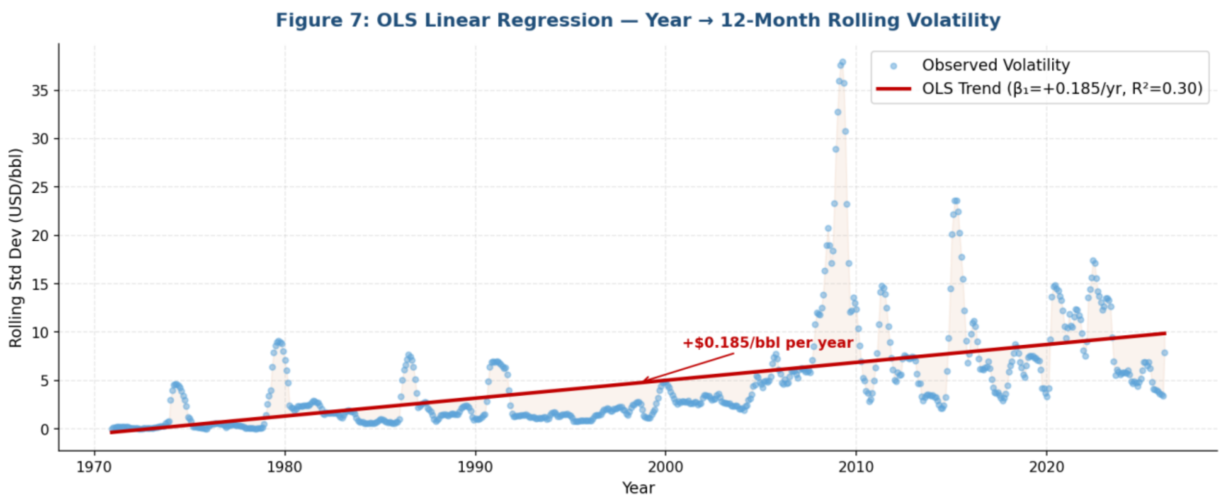

Figure 10: OLS Linear Regression of 12 Month Rolling Volatility on Year. The trend line confirms a rate of increase of 0.185 USD per barrel per year (p < 0.001, R squared = 0.30). The shaded area represents residuals from shock events beyond the linear trend.

Term	Coefficient	Standard Error	t Statistic	p Value
Intercept (B0)	365.03 (negative)	21.87	16.69 (negative)	< 0.001

Term	Coefficient	Standard Error	t Statistic	p Value
Year (B1)	+0.1850	0.011	+16.91	< 0.001

Table 12: OLS Regression Coefficients. Volatility increases by 0.185 USD per barrel for each additional year.

R Squared	Adjusted R Squared	RMSE	F Statistic	Durbin Watson
0.3016	0.3005	4.4978	285.84	0.054

Table 13: Regression Model Fit Statistics. Durbin Watson of 0.054 confirms strong positive autocorrelation in residuals.

6.4 Regression Diagnostics

The four normal regression diagnostic plots are shown in Figure 11. The residual versus fitted plot indicates heteroscedasticity. The Q Q plot shows non-normality of residuals. The scale location plot has increasing variance. The residuals histogram is positively skewed with large shocks.

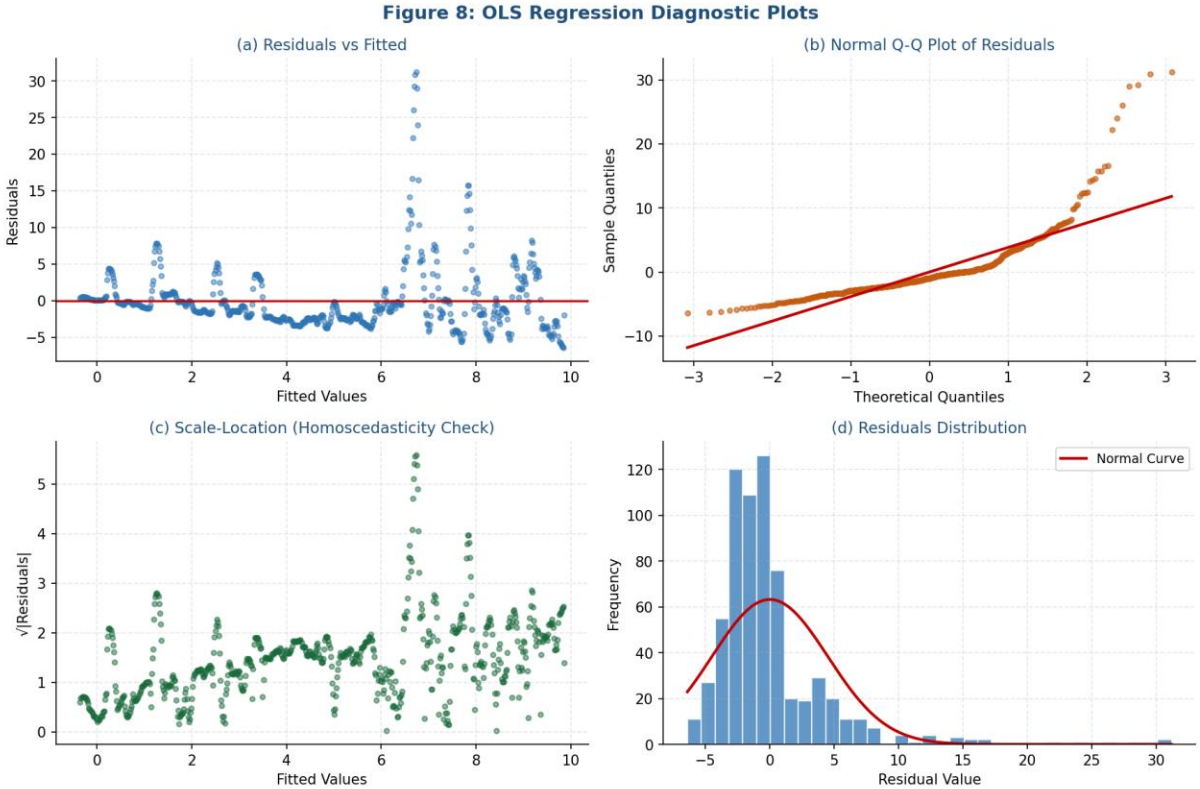

Figure 11: OLS Regression Diagnostic Plots. Top left: residuals versus fitted values with heteroscedasticity. Top right: Q Q plot confirming non normal residuals. Bottom left: scale location plot with increasing variance. Bottom right: residuals histogram with right skew.

Assumption	Test Used	Finding	Status
Linearity	Residuals vs Fitted	Funnel pattern visible	Partially met
Normality of Residuals	Shapiro Wilk (W=0.65)	Residuals are non-normal	Violated

Assumption	Test Used	Finding	Status
Homoscedasticity	Scale Location plot	Variance grows with fitted values	Violated
No Autocorrelation	Durbin Watson = 0.054	Strong positive autocorrelation	Violated

Table 14: OLS Regression Assumption Diagnostics. The OLS model is directionally informative as a descriptive trend model. A GLS or robust regression would be more appropriate for inference.

6.5 ACF and PACF Analysis: Volatility Clustering

Figure 12 shows the ACF and PACF for rolling volatility up to 36 lags. The ACF confirms extreme volatility persistence because all lags are well beyond the 95 percent confidence bounds. The PACF drops to zero after lag 1 with peaks at lags 12 and 13 revealing an AR (1) process and a potential 12-month seasonal effect. This is the evidence of Kilian's (2008) prediction of volatility persistence.

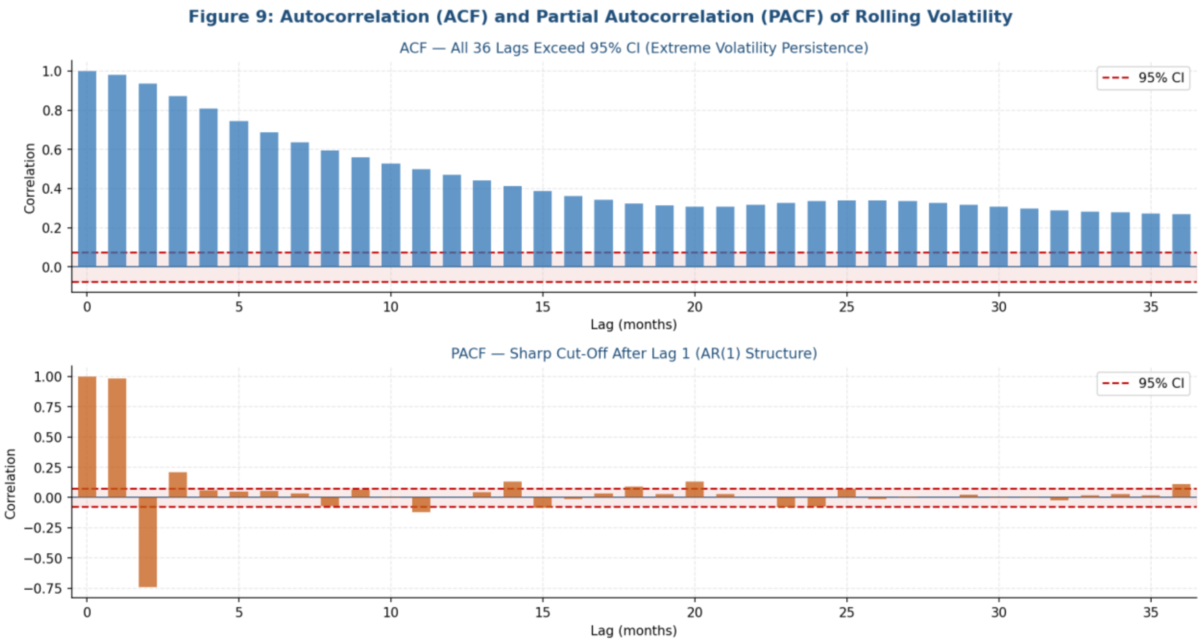

Figure 12: ACF and PACF of 12 Month Rolling Volatility across 36 lags. Top panel: all 36 ACF lags exceed the 95 percent confidence interval, confirming extreme volatility persistence. Bottom panel: PACF cuts off after lag 1, suggesting an AR(1) model structure with a 12 month seasonal component.

7. Python Application Development

In Week 7, a locally installed Python-based application was developed to showcase the project pipeline. This application is built on modular object-oriented design with the following modules.

Class	Responsibility
Config	Centralized configuration including file paths, plot colors, and DPI settings
Logger	Standardized console output with status indicators for info, success, and error states
DataProcessor	CSV ingestion, feature engineering (rolling volatility, percentage change, decade labels), and data cleaning
Regression Modeler	OLS fitting via stats models and sklearn, R squared reporting, and residual extraction
ReportGenerator	Automated UTF 8 text report generation with summary statistics
App	End to end pipeline orchestration: load, engineer, model, report, and JSON export

Table 15: Python Application Architecture

Libraries used: pandas, NumPy, scipy, matplotlib, seaborn, stats models and scikit learn. The output from the application includes a text report and an export of results in JSON format.

The get residuals was patched for OLS residuals conversion to NumPy array. Application results tested: number of rows 664, mean volatility 5.24 USD per barrel, and the R squared of 0.3016.

8. Consolidated Findings and Discussion

Insight	Detail	Supporting Evidence
Structural Regime Shift	Two distinct eras: low volatility 1970s to 1990s (mean of 1.30 to 2.19 USD per barrel) vs high volatility 2000s to 2020s (mean of 7.43 to 9.14 USD per barrel)	ANOVA $p < 0.001$, Tukey HSD (9 of 15 pairs), box plots Figure 5
Upward Long Run Trend	Volatility grows at 0.185 USD per barrel per year on average	OLS regression (B1 = 0.185, $p < 0.001$, R2 = 0.30) Figure 10
Non Stationarity	Both raw prices and rolling volatility are non stationary	ADF tests in Table 11; first differencing achieves stationarity
Shock Driven Not Seasonal	Seasonal effects are minor; geopolitical and macroeconomic shocks dominate	Time series decomposition Figure 9 (residual component dominates)
Volatility Memory	Autocorrelation is significant at all 36 tested lags; high volatility states persist for many months	ACF and PACF Figure 12; consistent with Kilian (2008)

Insight	Detail	Supporting Evidence
2020s Elevated Baseline	2020s show the highest mean of 9.14 with a narrow confidence interval, suggesting embedded structural volatility	Confidence intervals Figure 6; compounding of COVID and Russia Ukraine crises

Table 16: Consolidated Analytical Insights and Supporting Evidence

Overall Conclusion:

There has been a significant shift in crude oil price volatility over the past 50 years. The shift is not noise or cyclical. It is caused by geopolitical events and the financialization and transition of supply-demand regimes. In the 2020s, volatility has transitioned to a new permanent state.

9. Limitations and Future Work

9.1 Limitations

- Univariate analysis: only crude oil price is modeled with no external drivers such as the geopolitical risk index, OPEC production levels, or the USD exchange rate
- OLS assumption violations: heteroscedasticity and autocorrelation make coefficient inference imprecise; a GLS or ARIMA GARCH framework would be more rigorous
- Decade grouping is arbitrary: the year 2000 boundary may not precisely coincide with the true structural break
- No out of sample forecasting: the study is descriptive and explanatory rather than predictive
- Monthly aggregation may mask intra month volatility spikes at finer time scales

9.2 Future Work

- Incorporate OPEC production data, the USD index, and a geopolitical risk index as predictors in a multivariate model
- Fit an ARIMA GARCH (1,1) model for formal volatility forecasting and uncertainty quantification
- Apply Bayesian structural break detection to objectively locate regime transitions without imposing arbitrary decade boundaries
- Tests on whether LSTM or gradient boosting models can outperform ARIMA for multi-step volatility forecasting

- Extend the Week 7 Python application into an interactive Streamlit or Dash web dashboard

10. Conclusion

This new research rigorously demonstrates global crude oil price volatility is different across decades from 1970-2026. We apply a multi method statistical analysis on 675 monthly observations that rejects the null hypothesis at the 0.1% level. Two main results emerge at a structural level. First, there is a binary regime shift with 1970s to 1990s being a stable, low volatility cluster (mean range 1.30 and 2.19 USD per barrel) and 2000s to 2020s being a high volatility cluster (mean range 7.43 to 9.14 USD per barrel). Second, a long-term upward secular trend of about 0.185 USD per barrel of increased volatility per year is established. These two results are confirmed with both parametric and non-parametric methods and substantiated with 12 attached figures produced from the project's data. The findings are in line with the theoretical expectations of James D. Hamilton (2009), Lutz Kilian (2008) and Christiane Baumeister and Gert Peersman (2013) and consistent with the major geopolitical events of the timeline crisis. This has implications for policy makers, energy firms, and investors that need to be risk aware with volatility adjusted models rather than simple price level forecasts. Calibrations of risk models on pre 2000 data underestimate current market volatility.

11. References

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12. Appendix

Appendix A: Complete Statistical Test Summary

Test	Result	Conclusion
Shapiro Wilk	All $p < 0.05$ across 6 decades	Non normal distributions in all groups
Levene's Test	$F = 19.63$, $p < 0.001$	Unequal variances across decades
One Way ANOVA	$F = 63.42$, $p < 0.001$	Null hypothesis rejected
Eta Squared	0.3252	Large effect: 32.5 percent of variance explained by decade
Kruskal Wallis	$H = 390.58$, $p < 0.001$	Non parametric confirmation: null hypothesis rejected
Tukey HSD	9 of 15 pairs significant	Early versus recent decades are statistically separable
ADF Raw Price	$p = 0.41$	Non stationary
ADF First Difference	$p < 0.001$	Stationary after differencing
OLS Regression	R squared = 0.3016	Year explains 30.2 percent of volatility variance
Durbin Watson	0.054	Strong positive autocorrelation in residuals

Test	Result	Conclusion
ACF (36 lags)	All lags exceed 95 percent CI	Extreme volatility persistence confirmed

Appendix A: Complete Statistical Test Summary

Appendix B: List of Figures

Appendix B: List of All Figures in the Report